Team 39: Same Runs, Different Strategies

DAI Course Project

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1. Main Argument: Same Runs, Different Strategies

The same number of runs can be scored through different combinations of fours, sixes, and non-boundary scoring. We aim to investigate which scoring strategies are associated with batting success, how these relationships differ between teams, and how they vary with bowling conditions and innings phases. We will also examine how teams' scoring strategies change across seasons relative to the league average in each season.

1.1 Hypothesis A: Team-Specific Scoring Strategies

Do different IPL teams exhibit distinct scoring strategies, and which strategies are associated with success for each team? We will compare team-wise scoring compositions and their association with batting outcomes. Rather than examining boundary composition alone, we will investigate how these strategies behave under different bowling types, bowling difficulties, and innings phases.

Dimensions of analysis:

- **Bowling type:** Compare scoring patterns against pace and spin bowling to investigate whether different approaches are associated with success against each type.
- **Bowling difficulty:** Estimate bowling strength using historical performance metrics, such as economy rate and wicket-taking rate, and examine how scoring patterns vary against stronger and weaker bowling attacks.
- **Innings phase:** Compare scoring patterns during the powerplay (overs 1–6), middle overs (overs 7–15), and death overs (overs 16–20) to investigate phase-specific differences.

1.2 Hypothesis B: Team-Specific Strategies Across Seasons

How do a team's scoring strategies relative to the league average change across successful and unsuccessful seasons, and which patterns are associated with its better seasons? For each team and season, we will compare scoring characteristics with the corresponding league-wide averages for that season. This accounts for changes in scoring trends across IPL history. We will track relative differences in boundary scoring, non-boundary scoring, and dot-ball frequency across seasons to investigate which patterns are associated with better performance for individual teams.

We will further examine these season-wise differences across bowling types, bowling difficulties, and innings phases to investigate whether the strategies associated with success change over time.

1.3 Scoring Metrics and Statistical Analysis

We will derive metrics such as boundary-run share, four and six frequency, non-boundary scoring, and dot-ball percentage from ball-by-ball data. Depending on the question, we will compare these metrics alongside runs scored, wickets lost, and match outcomes. We will use suitable probability distributions, expected values, variances, visualisations, and statistical tests to investigate differences and associations.

1.4 Source Papers, Blogs, or Podcasts

- **Cricsheet – IPL match data:** Cricsheet IPL Match Dataset

1.5 Programming Languages, Tools, and Software

Python, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter Notebook, VS Code, Git, and GitHub.

1.6 Datasets

- **IPL ball-by-ball match data:** Cricsheet IPL Ball-by-Ball Dataset
- **JSON format documentation:** Cricsheet JSON Format Documentation

1.7 LLMs and AI Tools

- **ChatGPT conversation:** ChatGPT Discussion and Project Planning

2. Logistics

2.1 Project Deadlines

What	Start	End	Team Comments	Teaching Team Comments
Team Formation	Oct. 6, 2026	Oct. 9, 2026		
Brainstorming, Blackboarding	Oct. 9, 2026	Oct. 10, 2026	Considered various topics and ideas and listed them	
Project Proposal Due	Oct. 11, 2026	Oct. 11, 2026		Moodle submission; see course instructions.
Download Code and Build Baseline	Oct. 17, 2026	Oct. 12, 2026		
Disqualification from Teaching Team		Oct. 13, 2026		Hopefully not necessary.
Updated Proposal		Oct. 14, 2026		For disqualified proposals only.

What	Start	End	Team Comments	Teaching Team Comments
Data Collection	Oct. 12, 2026	Oct. 15, 2026		
Intermediate Checkpoint 1: Status Report for Submission 1	Oct. 20, 2026	Oct. 16, 2026	Refined the initial report into precise research questions and documented completed work.	Moodle submission: status report.
More Design, Brainstorming, Coding	Oct. 23, 2026	Oct. 27, 2026		
Freeze Coding and Result Generation	Oct. 27, 2026	Oct. 28, 2026		
Checkpoint 2	Nov. 3, 2026	Oct. 29, 2026		Submit the position paper as a PDF and the team presentation video.
Viva (if necessary)				A schedule will be announced if required.
Viva (if necessary)				

2.2 Repository and Collaboration Links

- **GitHub repository:** Same Runs, Different Strategies – GitHub Repository
- **Project spreadsheet and deadlines:** View Project Schedule and Deadlines